

The Epidemiology of Professional Ice Hockey Injuries in the National Hockey League

A Retrospective Analysis From 2012 to 2023

Derrick M. Knapik,^{*†‡} MD , Darius Sina Tartibi,[†] MD , Robert H. Brophy,^{†‡} MD ,
Matthew V. Smith,^{†‡} MD, and Matthew J. Matava,^{†‡} MD

Investigation performed at Washington University in St. Louis, St. Louis, Missouri, USA

Background: Epidemiological studies evaluating injuries in the National Hockey League (NHL) are primarily limited to publicly available data sources. The purpose of this study was to analyze all injuries in NHL athletes from the 2012-2013 through the 2022-2023 seasons (excluding the COVID-19–shortened seasons) to determine injury incidence, mechanism, location, and predictors of games and time lost due to injury.

Study Design: Descriptive epidemiology study; Level of evidence, 3.

Methods: All NHL player injuries and associated data from the 2012-2013 through 2022-2023 seasons, including the work stoppage–shortened 2012-2013 season but excluding the 2019-2020 and 2020-2021 seasons that were shortened due to the COVID-19 pandemic, were extracted from the Athlete Health Management System—the electronic medical record mandated for use by all NHL teams. Injury incidence density (IID) per 1000 athlete-exposures (AEs) was used for comparisons, and regression analysis was used to determine predictors of an injury resulting in ≥ 5 man-games lost (MGL).

Results: A total of 24,776 injuries were recorded. The IID was 50.2 injuries/1000 AEs in the preseason, 51.3 injuries/1000 AEs in the regular season, and 67.9 injuries/1000 AEs in the playoffs. The IID was highest in forwards (55.3 injuries/1000 AEs), while injuries were most common during the first period (38.2%). Puck-related injuries were the most common injury mechanism (25.9%), and the body regions most often injured were the head (21.2%) and hand (12.9%). Overall, a mean of 2.1 man-games (8.2 days) were lost per injury, with knee (4.2 MGL/injury; 17.2 days lost/injury) and shoulder (3.6 MGL/injury; 14.2 days lost/injury) injuries resulting in the greatest mean time loss per injury. Positive predictors of ≥ 5 MGL included knee (OR, 2.13) and abdominal (OR, 2.08) injuries, body checks (OR, 1.81), and the goaltender position (OR, 1.78).

Conclusion: From the 2012-2013 through 2022-2023 NHL seasons (excluding the 2019-2020 and 2020-2021 seasons shortened by the COVID-19 pandemic), the highest number of injuries occurred during the regular season, while the highest rate of injury occurred during the playoffs. The greatest incidence of injuries occurred in forwards, during the first period, due to puck-related causes. Injuries occurred most commonly to the head and hand, while injuries to the knee and shoulder accounted for the greatest mean number of man-games and days lost.

Keywords: NHL; hockey; injury; epidemiology

Ice hockey is a dynamic team sport characterized by the physical demands of speed, strength, and endurance, often resulting in high-speed collisions with opposing players and objects. Played within a confined rink bordered by boards and plexiglass, players can reach skating speeds exceeding 25 miles per hour (mph).⁷ The puck is a solid disc of vulcanized rubber 1 inch (2.54 cm) thick by 3 inches (7.6 cm) in diameter that weighs approximately 6 ounces

(170 g). The puck is propelled by sticks made of composite materials such as carbon fiber, fiberglass, or aluminum, and can travel at speeds well over 100 mph as it is shot toward a rectangular goal composed of tubular steel that is 72 inches (183 cm) wide by 48 inches (122 cm) high.^{27,28} Therefore, the unique playing environment in ice hockey inherently exposes players at all levels of play to a relatively high risk of injury.

While epidemiological studies have provided valuable insight into injury rates in elite ice hockey athletes, discrepancies in reported injury incidence exist due to variations in injury definitions and surveillance techniques.¹⁷ Reported injury rates in international Olympic and

professional men's ice hockey have ranged from 52.1 to 84 injuries per 1000 player-game hours.^{2,15,19,20,31} Notably, a study examining injury rates in the National Hockey League (NHL) from the 2006-2007 through 2011-2012 seasons identified an injury rate of 49.4 injuries per 1000 player-game hours.¹⁷ While these epidemiological studies have provided historical context, internal factors (ie, player fitness/conditioning, skill level, prior injuries, player age, etc) and external factors (ie, rule changes, equipment modification, training protocols, game strategy, player behavior, game schedules, travel, etc) may potentially affect both injury rates and types of injuries in the current NHL environment.

The purpose of this investigation was to determine the incidence and types of injuries in NHL athletes over the 2012-2013 through 2022-2023 seasons. We specifically evaluated the potential associations between injuries and player position, body region, period of game, time of season, and number of games and days lost for each specific injury as well as analyzed predictors of time loss secondary to injury. We hypothesized, based on prior epidemiological data from the NHL,¹⁷ that head, thigh and knee injuries would represent the most commonly reported body regions injured, with injury incidence greatest in the first period and toward the end of the regular season.

METHODS

The study protocol was preapproved by the Human Research Protection Office and Institutional Review Board at the senior author's (M.J.M.) institution as well as the NHL/NHL Players' Association (NHLPA) Joint Health & Safety Committee. A retrospective review was performed of all injuries reported in professional ice hockey players from all NHL teams from the 2012-2013 through the 2022-2023 seasons. The 2012-2013 season was shortened due to a labor dispute between the NHL and NHLPA, resulting in a reduced 48-game regular season with no preseason games. However, this shortened season was included in our analysis since standard league protocols remained in place regarding travel, alignment, games played per team, and the playoffs. Data from the 2019-2020 and 2020-2021 seasons were excluded due to the COVID-19 pandemic, which resulted in significant alterations to the standard NHL protocols regarding travel, league alignment, and season duration. Specifically, the

2019-2020 season was suspended on March 12, 2020, with teams playing 68 to 71 games (as opposed to the typical 82-game regular season), followed by playoffs in which all teams played in 1 of 2 sites (bubbles): Edmonton, Alberta, or Toronto, Ontario, with finals in September 2020. During the 2020-2021 season, the regular season was reduced to 56 games, followed by a postseason playoff.

All injuries were extracted from an electronic medical record (EMR) (Athlete Health Management System [AHMS]), which is a league-mandated system used by all NHL teams, with data entered by NHL athletic trainers, physical therapists, and team physicians. All reported injuries occurred during hockey-related activities, including preseason, regular season, and playoff games. Practice-related injuries were not included within the AHMS database. Data were collected pertaining to the season, time of season (preseason, regular season, and playoffs), occurrence of injury during pregame or in-game activity, game period (first period, second period, third period, and overtime), player position group (goaltender, defenseman, and forward), diagnosis, body region (abdomen, ankle, chest, foot, hand, head, hip/groin, lower leg, knee, back [cervical, thoracic, or lumbar spine], neck, shoulder, thigh, upper arm, and other [including nonmusculoskeletal injuries]), mechanism of injury (hit by puck, intentional player contact [eg, body check], contact with the environment [ie, boards, glass, ice, goal, etc], fighting, noncontact, and unknown), regular season games lost, total games lost (including preseason, regular season, and playoffs), total games lost over multiple seasons (if applicable), and total days lost during the season.

Data were extracted to a Microsoft Excel spreadsheet. Descriptive statistics and nonparametric testing were performed to analyze the number of injuries occurring by each season, time of season (preseason, regular season, and playoffs), game period, player position group, body region, and injury mechanism. The mean numbers of overall games and days lost per injury were also calculated. Athlete-exposures (AEs) were estimates for overall player-hours based on the number of players per team (18 skaters and 1 goaltender)⁴ and the total number of games played per season across all teams. Injury incidence density (IID) was derived from the number of injuries per 1000 AEs. Injury density ratio (IDR) was derived from the relative risk ratio between 2 IIDs.¹⁷ Logistic regression (adjusting for time of season, player position group, body region, game period, and injury mechanism category) was used to identify potential positive and negative predictors

*Address correspondence to Derrick M. Knapik, MD, Washington University in St. Louis, School of Medicine, 14532 S Outer Forty Dr, Chesterfield, MO 63017, USA (email: knapik@wustl.edu).

[†]Department of Orthopaedic Surgery, Division of Sports Medicine, Washington University in St. Louis, St. Louis, Missouri, USA.

[‡]Washington University in St. Louis, School of Medicine, St. Louis, Missouri, USA.

Submitted August 30, 2025; accepted December 15, 2025.

One or more of the authors has declared the following potential conflict of interest or source of funding: D.M.K. reported grant support from Arthrex, and education support from Smith & Nephew and Elite Orthopaedics. R.H.B. reported speaking fees/services other than consulting, hospitality payments, and education support from Arthrex; consulting fees and hospitality payments from Sanofi; and education support from Elite Orthopaedics, Smith & Nephew, and BREGG. M.V.S. reported serving as a speaker for Elite Orthopaedics, and consulting for Arthrex and Flexion Therapeutics. M.J.M. reported educational activities/travel and lodging from Arthrex; educational activities for Elite Orthopaedics/Apollo Orthopaedics, and consulting for Pacira Pharmaceuticals and Heron Therapeutics. AOSM checks author disclosures against the Open Payments Database (OPD). AOSM has not conducted an independent investigation on the OPD and disclaims any liability or responsibility relating thereto.

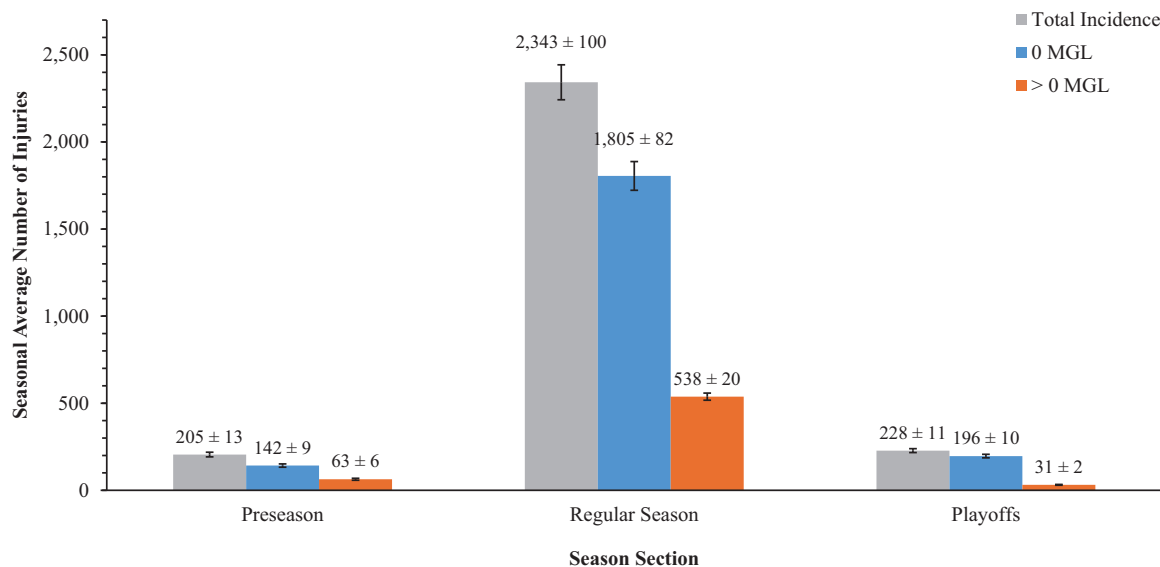


Figure 1. Mean number of injuries per season by preseason, regular season, and playoffs. Total incidence is the mean number of injuries per season including all injuries regardless of number of man-games lost (MGL). Zero MGL is the mean number of injuries per season including only injuries resulting in no MGL, and >0 MGL is the mean number of injuries per season including only injuries resulting in at least 1 MGL. Data are presented as mean $\pm$ standard error

for ≥ 5 man-games lost (MGL). The 5 MGL cutoff value was used to derive the odds ratio and 95% confidence interval. Five MGL has been previously reported to correspond to a mean of 10 days of time loss, which is considered a moderate or severe injury using prior hockey injury classification schemes.^{4,17,19,25}

RESULTS

A total of 24,776 unique injuries were reported in the NHL during the study period (Appendix 1, available in the online version of this article). The median number of injuries per season was 2764 (interquartile range [IQR], 2732-2951 injuries). Only 0.7% (183/24,776) of the injuries were classified as reinjuries.

Injuries by Season and Time of Season

Most injuries (85.1%; 21,084/24,776) occurred during regular season games, while 8.3% (2050/24,776) occurred in the playoffs and 6.6% (1642/24,776) occurred in the preseason. The mean number of injuries occurring per season by preseason, regular season, and playoffs is displayed in Figure 1. The numbers of zero time-loss and nonzero time-loss injuries occurring by time of season are shown in Figure 2. The overall IID was 50.2 injuries/1000 AEs in the preseason, 51.3 injuries/1000 AEs in the regular season, and 67.9 injuries/1000 AEs in the playoffs (Table 1). The overall IID, irrespective of time loss, was significantly greater in the playoffs compared to the regular season (IDR, 1.32; 95% CI, 1.28-1.37). For nonzero time-loss injuries, the IID was 15.5 injuries/1000 AEs in preseason, 11.8

injuries/11,000 AEs during the regular season, and 9.3 injuries/1000 AEs during the playoffs. The overall IID, for time-loss injuries only, was significantly lower in the playoffs compared to the regular season (IDR, 0.79; 95% CI, 0.67-0.91). During the regular season, there was a significantly higher IID in the first 4 seasons (2012-2013 through 2015-2016) when compared to the latter 5 seasons (2016-2017 through 2022-2023) (IDR, 1.16; 95% CI, 1.14-1.19). There was no significant difference in IID during the playoffs between the first 4 and latter 5 seasons (IDR, 1.01; 95% CI, 0.92-1.09) (Table 2).

Injuries by Player Position Group

Overall, 61.2% (15,162/24,776) of injuries were recorded in forwards, 34.9% (8642/24,776) in defensemen, and 3.9% (972/24,776) in goaltenders. When adjusting for the number of each player position per team (ie, forwards as 3/6, defensemen as 2/6, and goaltender as 1/6), the adjusted number of injuries per player position was 5054 in forwards, 4321 in defensemen, and 972 in goaltenders (Figure 3).

The IID was higher in forwards (55.3 injuries/1000 AEs) as compared to defensemen (IDR, 1.12; 95% CI, 1.09-1.14) and goaltenders (IDR, 1.42; 95% CI, 1.35-1.48), while the IID was higher in defensemen when compared to goaltenders (IDR, 1.27; 95% CI, 1.21-1.34).

Injuries by Game Period

Most injuries occurred during the first period (38.2%; 9470/24,776), followed by the second period (29.0%; 7185/24,776) and third period (26.2%; 6483/24,776). Only 1.1% (270/

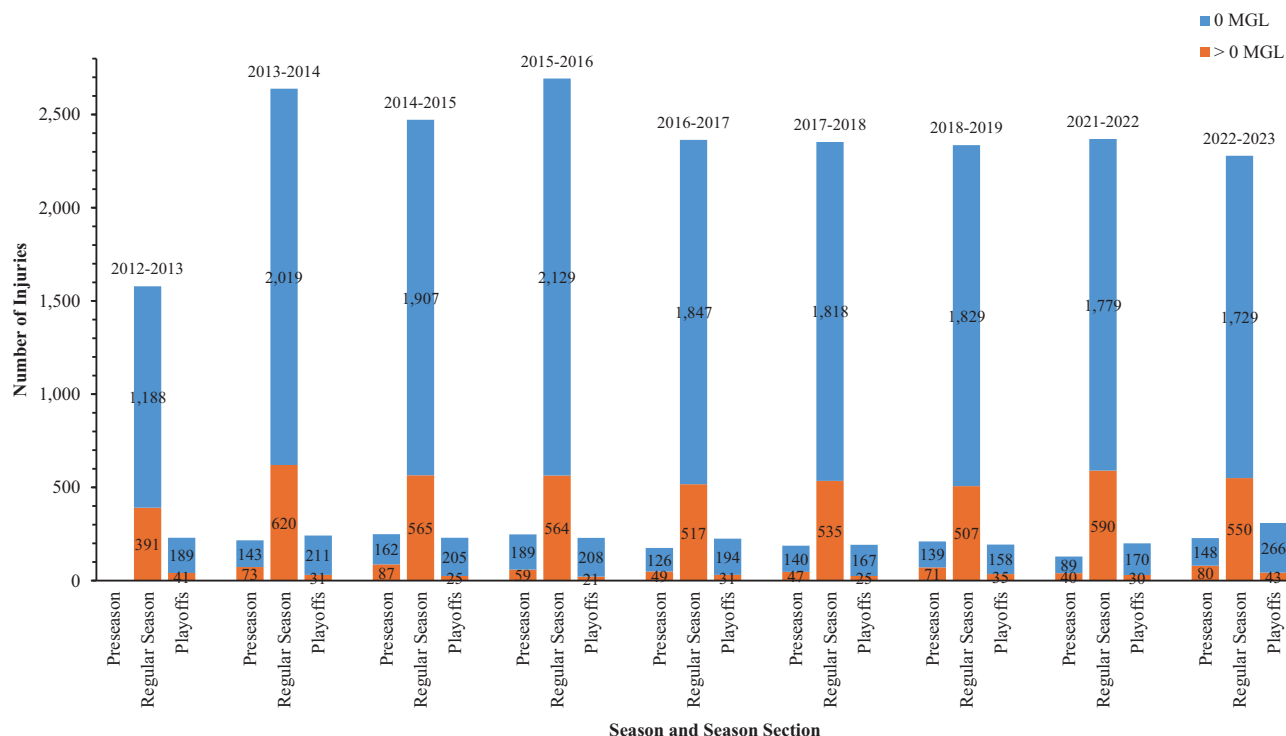


Figure 2. Number of zero time-loss and nonzero time-loss injuries by season and season section. Zero man-games lost (MGL) is the number of injuries resulting in no MGL, and >0 MGL is the number of injuries resulting in at least 1 MGL.

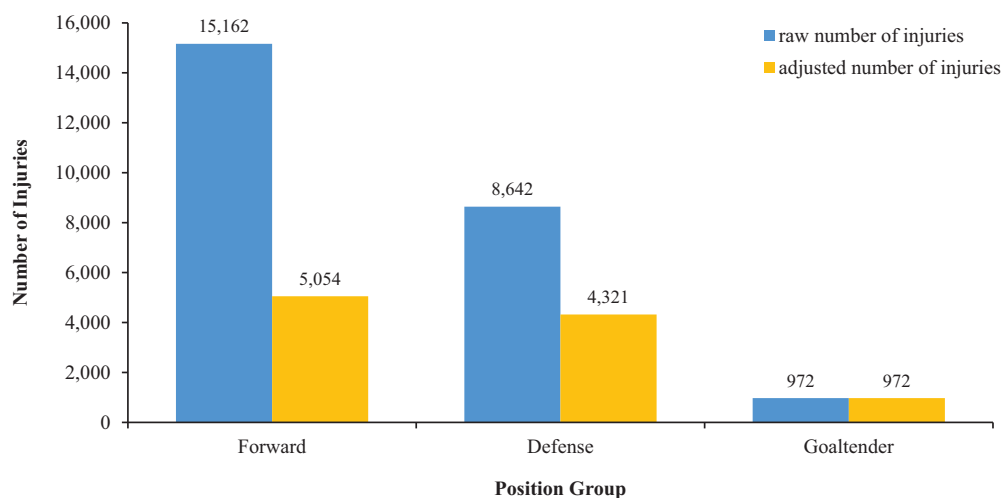


Figure 3. Total number of player injuries by player position group (forward, defense, or goaltender) from the 2012-2013 through the 2022-2023 seasons, excluding the 2 COVID seasons (2019-2020 and 2020-2021). Forward includes the Right Wing, Left Wing, Forward, and Center positions. The raw number of injuries is the total number of injuries occurring in the position group. The adjusted number of injuries is the number of injuries on a per-player basis adjusting for the number of positions per team (forwards, 3/6; defense, 2/6; goaltender, 1/6).

24,776) of injuries were recorded during pregame warm-up, and 0.8% (191/24,776) of injuries occurred during over-time. The game period was unknown in 4.8% (1177/24,776) of injuries. A full breakdown of zero time-loss and nonzero time-loss injuries by game period is displayed in Figure 4.

The IID was higher in the first period compared to the second (IDR, 1.32; 95% CI, 1.29-1.35) and third (IDR, 1.46; 95% CI, 1.42-1.50) periods, while the IID was higher in the second period compared to the third period (IDR, 1.11; 95% CI, 1.08-1.14).

TABLE 1
IID by Season and Season Section^a

	2012- 2013	2013- 2014	2014- 2015	2015- 2016	2016- 2017	2017- 2018	2018- 2019	2021- 2022	2022- 2023	Overall
Preseason										
Total injuries	NA	216	249	248	175	187	210	129	228	1642
Zero time-loss injuries		143	162	189	126	140	139	89	148	1136
Nonzero time-loss injuries		73	87	59	49	47	71	40	80	506
Estimated AEs	NA	3990	3990	3990	3990	4123	4123	4256	4256	32,718
Total IID (per 1000 AEs)		54.1	62.4	62.2	43.9	45.4	50.9	30.3	53.6	50.2
Zero time-loss IID	NA	35.8	40.6	47.4	31.6	34.0	33.7	20.9	34.8	34.7
Nonzero time-loss IID		18.3		14.8	12.3	11.4	17.2	9.4	18.8	15.5
Regular season										
Total injuries	1579	2639	2472	2693	2364	2353	2336	2369	2279	21,084
Zero time-loss injuries	1188	2019	1907	2129	1847	1818	1829	1779	1729	16,245
Nonzero time-loss injuries	391	620	565	564	517	535	507	590	550	4839
Estimated AEs	27,360	46,740	46,740	46,740	46,740	48,298	48,298	49,856	49,856	410,628
Total IID (per 1000 AEs)	57.7	56.5	52.9	57.6	50.6	48.7	48.4	47.5	45.7	51.3
Zero time-loss IID	43.4	43.2	40.8	45.5	39.5	37.6	37.9	35.7	34.7	39.6
Nonzero time-loss IID	14.3	13.3	12.1	12.1	11.1	11.1	10.5	11.8	11.0	11.8
Playoffs										
Total injuries	230	242	230	229	225	192	193	200	309	2050
Zero time-loss injuries	189	211	205	208	194	167	158	170	266	1768
Nonzero time-loss injuries	41	31	25	21	31	25	35	30	43	282
Estimated AEs	3268	3534	3382	3458	3306	3192	3306	3382	3344	30,172
Total IID (per 1000 AEs)	70.4	68.5	68.0	66.2	68.1	60.2	58.4	59.1	92.4	67.9
Zero time-loss IID	57.8	59.7	60.6	60.2	58.7	52.3	47.8	50.3	79.5	58.6
Nonzero time-loss IID	12.5	8.8	7.4	6.1	9.4	7.8	10.6	8.9	12.9	9.3

^aAE, athlete-exposure; IID, injury incidence density; NA, not applicable (2012-2013 was a lockout season in which the regular season was shortened and no preseason games were played).

TABLE 2
Injury Incidence Density in the First 4
Seasons as Compared to the Last 5 Seasons^a

	2012-2016	2016-2023	Overall
Preseason			
Injuries	713	929	1642
Estimated AEs	11,970	20,748	32,718
IID (per 1000 AEs)	59.6	44.8	50.2
Regular season			
Injuries	9383	11,701	21,084
Estimated AEs	167,580	243,048	410,628
IID (per 1000 AEs)	56.0	48.1	51.3
Playoffs			
Injuries	931	1119	2050
Estimated AEs	13,642	16,530	30,172
IID (per 1000 AEs)	68.2	67.7	67.9

^aAE, athlete-exposure; IID, injury incidence density; NA, not applicable (2012-2013 was a lockout season in which the regular season was shortened and no preseason games were played).

Injuries by Mechanism

Puck-related injuries were the most common injury mechanism (25.9%; 6413/24,776), followed by body checks (17.6%; 4364/24,776) and incidental contact with another player/equipment (15.6%; 3858/24,776) (Figure 5). Injuries

related to fighting accounted for only 3.2% (785/24,776) of all injuries during the period analyzed. Full data on zero and nonzero time-loss injuries as well as the percentage of injury mechanism per season over the course of the study period are displayed in Figures 5 and 6.

When evaluating injury mechanism based on time of season, puck-related injuries accounted for the highest prevalence of injuries during the preseason (20.1%; 332/1649), followed by body check-related injuries (17.1%; 282/1649) (Table 3). During the regular season, puck-related injuries once again accounted for the greatest prevalence of reported injuries (26.7%, 5638/21,077), while during the playoffs, body check-related injuries were the most commonly reported (23.2%, 475/2050). When assessing injury mechanism based on game period, puck-related injuries accounted for the highest prevalence of injuries during pregame warm-up (27.4%; 74/270) and first period (24.5%; 2323/9470), second period (28%; 2009/7185), and third period (29%; 1883/6483) play, while puck-related injuries were reported in identical prevalence with noncontact injuries (both 23.6%; 45/191) during overtime play (Table 4).

Injuries by Body Region

Head injuries accounted for the highest percentage of injuries (21.2%; 5259/24,776), followed by hand (12.9%; 3189/24,776), hip/groin (9.6%; 2384/24,776), and knee (8.7%;

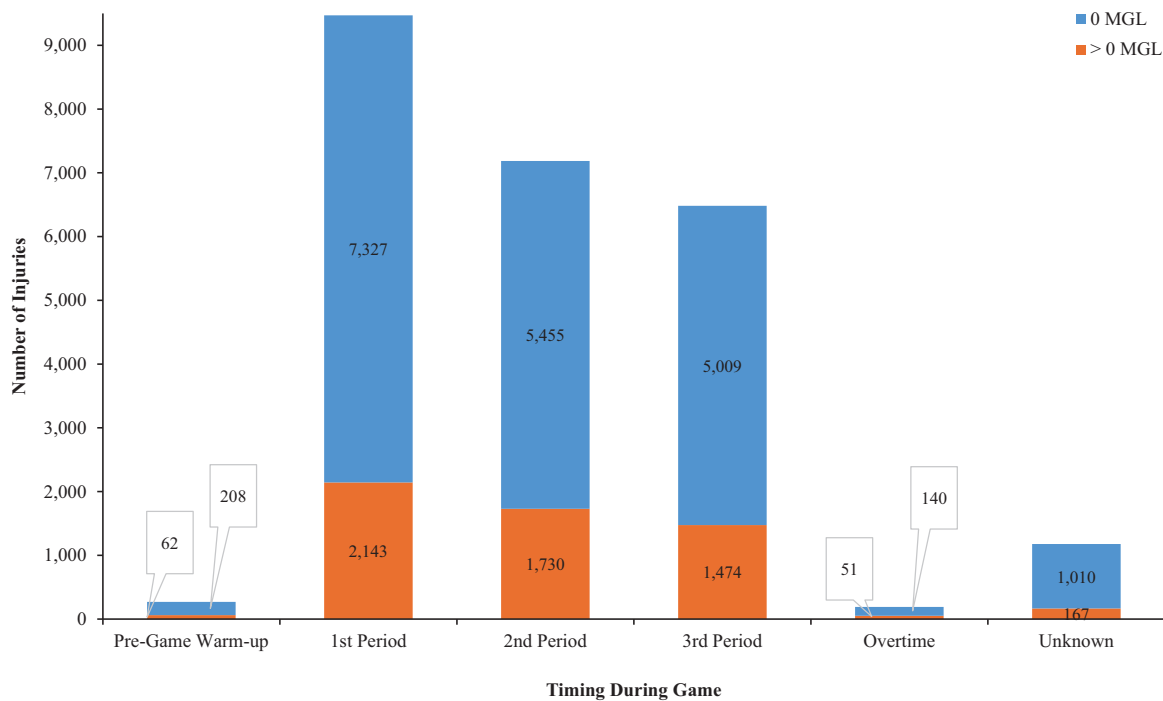


Figure 4. Total number of player injuries by timing during the game (pregame, first period, second period, third period, or overtime) from the 2012-2013 through the 2022-2023 seasons, excluding the 2 COVID seasons (2019-2020 and 2020-2021), resulting in zero man-games lost (MGL) versus nonzero man-games time lost.

2155/24,776) injuries. There were 5627 total injuries resulting in any number (>0) of MGL. Head injuries were the most prevalent (19.3%; 1084/5627), followed by hip/groin (14.9%; 839/5627) and knee (12.7%; 714/5627) injuries. There were 8634 total injuries resulting in any number of days lost. Similarly, head (15.8%; 1362/8634), hip/groin (13.4%; 1153/8634), and knee (11.6%; 1001/8634) injuries were the most prevalent (Figure 7). The most common injury diagnoses to the head were facial lacerations (9.1%; 2256/24,776) and concussions (3.9%; 954/24,776) (Appendix 1, available online).

A total of 52,377 man-games (203,091 days) were lost during the study period due to injury (Table 5). Overall, a mean of 2.1 man-games (8.2 days) were lost per injury, with knee (4.2 MGL/injury; 17.2 days lost/injury), shoulder (3.6 MGL/injury; 14.2 days lost/injury), and ankle (2.9 MGL/injury; 11.0 days lost/injury) injuries resulting in the largest mean time loss per injury. Concussions specifically resulted in a mean of 7.7 games lost per injury (Appendix 1, available online).

Excluding injuries with no MGL, there was an overall mean of 9.3 MGL/injury, with lower leg (14.6 MGL/injury), knee (12.6 MGL/injury), and ankle (11.1 MGL/injury) injuries resulting in the highest mean MGL per injury. Excluding injuries with no days lost, there was an overall mean of 23.5 days lost per injury, with knee (36.9 days lost/injury), shoulder (31.7 days lost/injury), and ankle (26.8 days lost/injury) injuries resulting in the highest mean days lost per injury.

Injury Risk Factors

Positive predictors of sustaining an injury resulting in ≥ 5 MGL included knee (OR, 2.13; 95% CI, 1.90-2.39), abdominal (OR, 2.08; 95% CI, 1.63-2.65), and ankle (OR, 2.00; 95% CI, 1.73-2.32) injuries; body checks (OR, 1.81; 95% CI, 1.66-1.99) and noncontact injuries (OR, 1.40; 95% CI, 1.23-1.58); the goaltender position (OR, 1.78; 95% CI, 1.59-1.99); and injury during the preseason (OR, 1.35; 95% CI, 1.19-1.53) and regular season (OR, 1.57; 95% CI, 1.43-1.71) (Table 6).

Negative predictors of sustaining an injury resulting in ≥ 5 MGL included neck (OR, 0.32; 95% CI, 0.24-0.42), thigh (OR, 0.55; 95% CI, 0.43-0.69), and upper arm (OR, 0.58; 95% CI, 0.47-0.73) injuries; puck injuries (OR, 0.52; 95% CI, 0.46-0.58); and injuries during the playoffs (OR, 0.47; 95% CI, 0.41-0.55).

When examining injuries resulting in ≥ 2 MGL, the predictors listed above remained largely unchanged except for the inclusion of 2 additional negative predictors: hand injuries (OR, 0.87; 95% CI, 0.78-0.97) and incidental contact with another player (OR, 0.86; 95% CI, 0.79-0.94).

DISCUSSION

Injuries in the NHL primarily occurred during the regular season (85.1%), with a greater IID during the course of the first 4 seasons (2012-2013 through 2015-2016) when

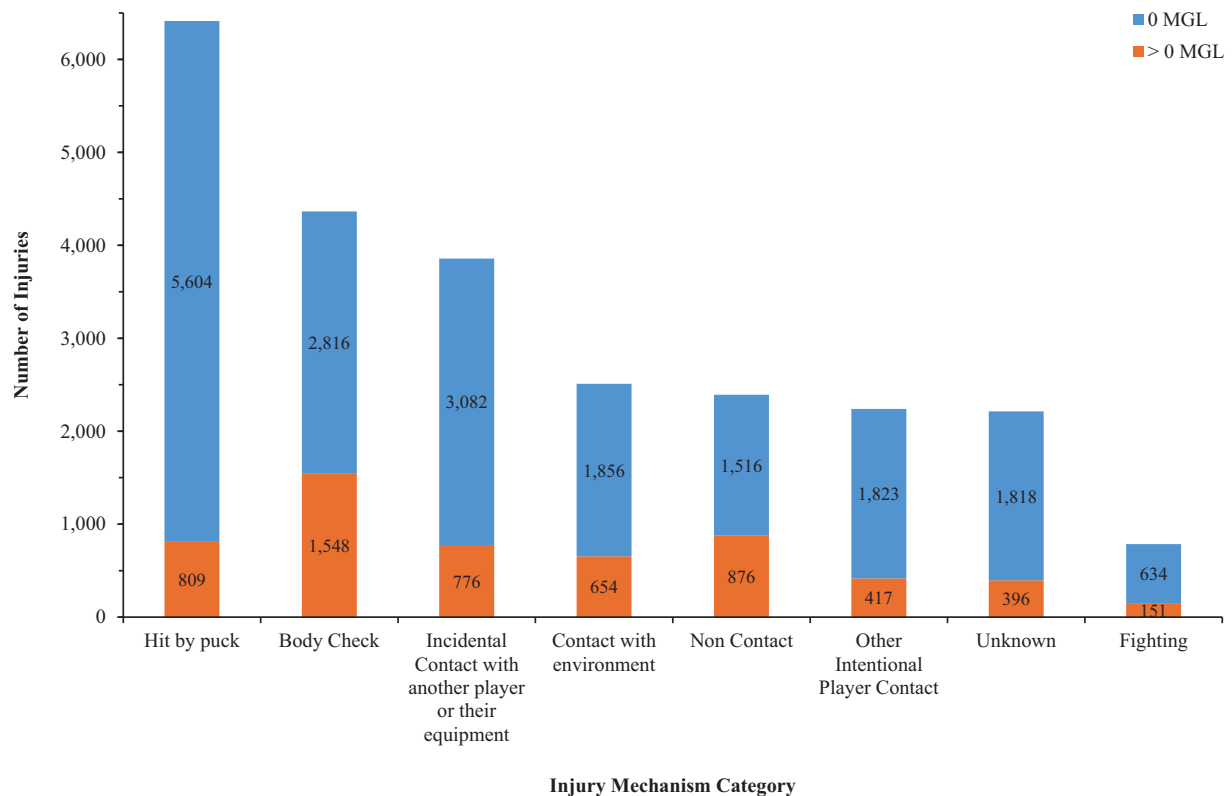


Figure 5. Total number of player injuries by category of injury from the 2012-2013 through the 2022-2023 seasons, excluding the 2 COVID seasons (2019-2020 and 2020-2021). The injury category was unknown for 2214 injuries.

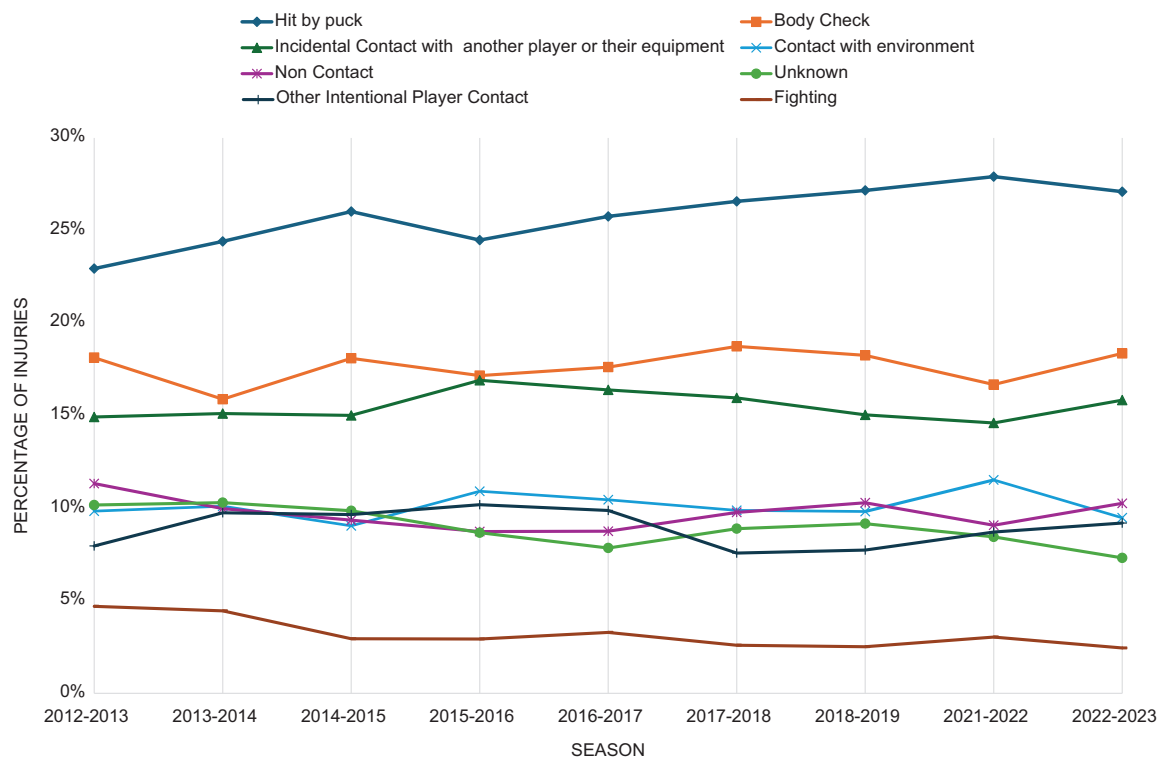


Figure 6. Percentage of overall injuries per season by injury mechanism over time.

TABLE 3
Prevalence of Injury Mechanism by Time of Season

Injury Mechanism	Preseason	Regular Season	Playoffs	Overall
Hit by puck	20.1% (332/1649)	26.7% (5638/21,077)	21.6% (443/2050)	25.9% (6413/24,776)
Body check	17.1% (282/1649)	17.1% (3607/21,077)	23.2% (475/2050)	17.6% (4364/24,776)
Incidental contact with another player or their equipment	16.3% (269/1649)	15.6% (3288/21,077)	14.7% (301/2050)	15.6% (3858/24,776)
Contact with environment	10.8% (178/1649)	10.0% (2112/21,077)	10.7% (220/2050)	10.1% (2510/24,776)
Noncontact	15.3% (253/1649)	9.4% (1975/21,077)	8.0% (164/2050)	9.7% (2392/24,776)
Other intentional player contact	7.0% (115/1649)	9.0% (1897/21,077)	11.1% (228/2050)	9.0% (2240/24,776)
Unknown	8.4% (138/1649)	8.9% (1878/21,077)	9.7% (198/2050)	8.9% (2214/24,776)
Fighting	5.0% (82/1649)	3.2% (682/21,077)	1.0% (21/2050)	3.2% (785/24,776)
Overall	100.0% (1649/1649)	100.0% (21,077/21,077)	100.0% (2050/2050)	100.0% (24,776/24,776)

TABLE 4
Prevalence of Injury Mechanism by Game Period

Injury Mechanism	Pre Game Warm-up	1st Period	2nd Period	3rd Period	Overtime	Unknown	Overall
Hit by puck	27.4% (74/270)	24.5% (2323/9470)	28.0% (2009/7185)	29.0% (1883/6483)	23.6% (45/191)	6.7% (79/1177)	25.9% (6413/24,776)
Body check	0.4% (1/270)	18.5% (1748/9470)	19.5% (1399/7185)	17.2% (1114/6483)	10.5% (20/191)	7.0% (82/1177)	17.6% (4364/24,776)
Incidental contact with another player or their equipment	4.4% (12/270)	16.3% (1540/9470)	16.6% (1190/7185)	16.0% (1036/6483)	14.7% (28/191)	4.4% (52/1177)	15.6% (3858/24,776)
Contact with environment	2.6% (7/270)	9.9% (940/9470)	10.9% (780/7185)	10.8% (698/6483)	9.9% (19/191)	5.6% (66/1177)	10.1% (2510/24,776)
Noncontact	42.6% (115/270)	9.8% (928/9470)	7.6% (547/7185)	8.6% (558/6483)	23.6% (45/191)	16.9% (199/1177)	9.7% (2392/24,776)
Other intentional player contact	0.0% (0/270)	8.7% (821/9470)	9.4% (673/7185)	10.8% (698/6483)	9.9% (19/191)	2.5% (29/1177)	9.0% (2240/24,776)
Unknown	22.6% (61/270)	8.5% (809/9470)	4.6% (329/7185)	5.2% (334/6483)	7.3% (14/191)	56.7% (667/1177)	8.9% (2214/24,776)
Fighting	0.0% (0/270)	3.8% (361/9470)	3.6% (258/7185)	2.5% (162/6483)	0.5% (1/191)	0.3% (3/1177)	3.2% (785/24,776)
Overall	100.0% (270/270)	100.0% (9470/9470)	100.0% (7185/7185)	100.0% (6483/6483)	100.0% (191/191)	100.0% (1177/1177)	100.0% (24,776/24,776)

compared to the subsequent 5 seasons (2016-2017 through 2022-2023). The greatest incidence of injuries occurred in forwards, during the first period, due to puck-related causes and body checks, with the overall highest IID occurring during the playoffs. From an anatomic perspective, injuries occurred most commonly to the head, hand, and hip/groin. Of the total 52,377 MGL, injuries to the knee, shoulder, and ankle accounted for the highest mean number of games lost.

McKay et al¹⁷ evaluated the IID from the 2006-2007 through 2011-2012 NHL seasons using the NHL standardized electronic injury surveillance. They found that of the 4368 regular season and 230 playoff injuries reported with time lost, the IID per 1000 AEs was greater in the regular season (15.6) relative to the playoffs (11.7).¹⁷ This investigation found the overall time-loss IID to be 11.8 in the regular season and 9.3 in the playoffs. Given a relative similarity in estimated AEs between our study and that of McKay, the decrease in IID is primarily explained by the decrease in the number of reported injuries.

Advances in player safety regulations mandated by the NHL likely explain at least some of the reduction in injuries resulting in time lost. Stricter criteria for suspension from game for misconduct were implemented in the 2014-2015 season (Rule 23.6).^{22,24} Game misconduct included boarding, charging, checking from behind, clipping, elbowing, head butting, interference, and kneeing. Because of

the relatively high prevalence of time-loss injuries resulting from contact with other players as well as injuries to the head and knee, these enforcements may have contributed to reducing the overall injury incidence in our study. The 2010-2011 season introduced Rule 48, which made some checks to the head illegal.^{22,24} In a retrospective study, Hutchison et al¹² reported a decreased incidence of concussions in the NHL after the introduction of Rule 48. The NHL protocol for concussion evaluation and management has evolved with the field since 2011.²³ Furthermore, the 2013-2014 season saw the implementation of mandatory visors for players with <25 games of experience (Rule 9.7) as well as a minor penalty for players who removed their helmets before an in-game fight (Rule 46.6).^{21,24} Because head injuries comprised the highest incidence of time-loss injuries, these safety measures may have further contributed to the decreased incidence of facial and head injuries and, subsequently, the overall IID reported during this study period.

The AHMS was implemented by the NHL in 2006, coinciding with the start date of data collection used by McKay et al.¹⁷ Prior investigations^{1,11} have reported that new EMR systems may take months to years for provider adaptation and accurate usage, potentially leading to underreporting during AHMS implementation. The extent to which injuries were potentially underreported cannot be objectively measured.^{16,17} Furthermore, the IID over the

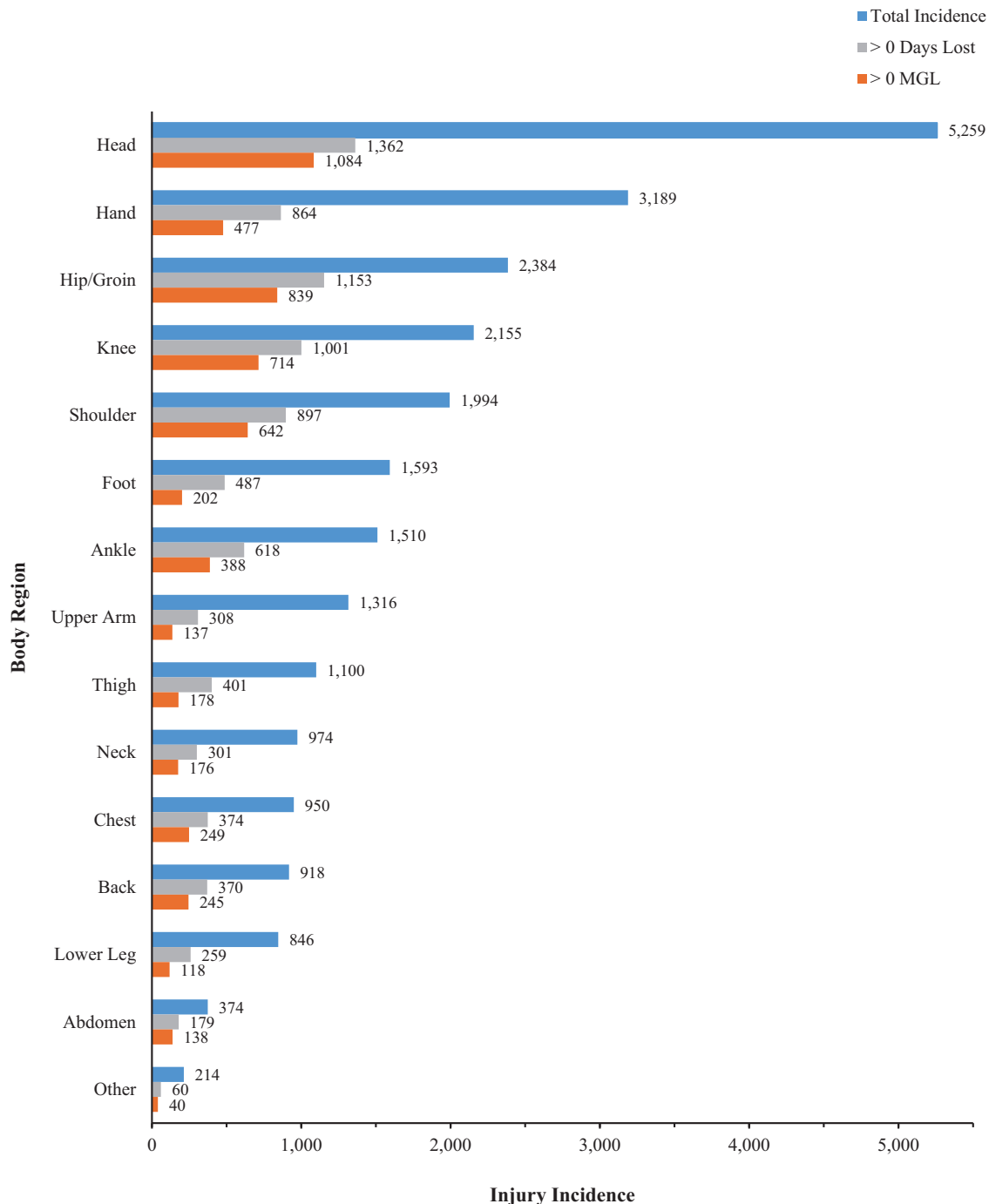


Figure 7. Incidence of player injuries resulting in time loss based on body region from the 2012-2013 through the 2022-2023 seasons, excluding the 2 COVID seasons (2019-2020 and 2020-2021). MGL, man-games lost.

course of the first 4 regular seasons (2012-2013 through 2015-2016) was greater when compared to the subsequent 5 seasons (2016-2017 through 2022-2023). This may be explained by a rule change in the 2015-2016 season that required regular season overtime games to be played with 3 skaters and 1 goaltender, down from the previous 4 skaters and 1 goaltender. This may also explain why a similar decrease in injury density was not observed in

the playoffs, which maintained the standard 5 skaters and 1 goaltender for an unlimited number of full 20-minute sudden-death overtime periods; however, as no direct causality can be inferred from the present study, further investigation is necessary.¹⁰

Despite the overall greater incidence of total injuries during the regular season, the overall IID was the highest during the playoffs, whereas the nonzero time-loss IID was

TABLE 5
Games and Days Missed by Body Region Injured^a

Body Region	Injury Incidence (%)	Injury Incidence, Nonzero (%)	Regular Season Games Lost	All Season Games Lost	Total Games Lost (%)	Total Days Lost (%)	Mean Games Lost Per Injury	Mean Games Lost Per Injury, Nonzero	Mean Days Lost Per Injury	Mean Days Lost Per Injury, Nonzero
Head	5259 (21.2)	1084 (19.3)	6997	7638	8918 (17.0)	31,777 (15.6)	1.7	8.2	6.0	23.3
Hand	3189 (12.9)	477 (8.5)	4400	4789	4952 (9.5)	20,185 (9.9)	1.6	10.4	6.3	23.4
Hip/groin	2384 (9.6)	839 (14.9)	5134	5700	5986 (11.4)	22,113 (10.9)	2.5	7.1	9.3	19.2
Knee	2155 (8.7)	714 (12.7)	7554	8182	8972 (17.1)	36,981 (18.2)	4.2	12.6	17.2	36.9
Shoulder	1994 (8.0)	642 (11.4)	6004	6496	7093 (13.5)	28,404 (14.0)	3.6	11.0	14.2	31.7
Foot	1593 (6.4)	202 (3.6)	1476	1547	1646 (3.1)	5450 (2.7)	1.0	8.1	3.4	11.2
Ankle	1510 (6.1)	388 (6.9)	3818	4147	4309 (8.2)	16,585 (8.2)	2.9	11.1	11.0	26.8
Upper arm	1316 (5.3)	137 (2.4)	1243	1358	1368 (2.6)	5519 (2.7)	1.0	10.0	4.2	17.9
Thigh	1100 (4.4)	178 (3.2)	958	1102	1135 (2.2)	5064 (2.5)	1.0	6.4	4.6	12.6
Neck	974 (3.9)	176 (3.1)	880	978	1210 (2.3)	4970 (2.4)	1.2	6.9	5.1	16.5
Chest	950 (3.8)	249 (4.4)	1449	1583	1684 (3.2)	6128 (3.0)	1.8	6.8	6.5	16.4
Back	918 (3.7)	245 (4.4)	1486	1630	2040 (3.9)	8280 (4.1)	2.2	8.3	9.0	22.4
Lower leg	846 (3.4)	118 (2.1)	1583	1710	1718 (3.3)	6885 (3.4)	2.0	14.6	8.1	26.6
Abdomen	374 (1.5)	138 (2.5)	971	1028	1028 (2.0)	3311 (1.6)	2.7	7.4	8.9	18.5
Other	214 (0.9)	40 (0.7)	283	303	318 (0.6)	1439 (0.7)	1.5	8.0	6.7	24.0
Overall	24,776 (100.0)	5627 (100.0)	44,236	48,191	52,377 (100.0)	203,091 (100.0)	2.1	9.3	8.2	23.5

^aInjury incidence is the percentage of total injuries by body region injured. Games lost is the total number of games lost during regular season, during the entire season, and across all seasons due to a specific body region being injured. Total days lost is the total number of days missed due to specific body region being injured. Mean games lost and days lost per injury are the mean number of games and days missed per injury of the specified body region. Nonzero is the mean time loss per injury excluding injuries in which no games or days were missed.

highest during preseason, followed by the regular season and playoffs (Table 1). This finding aligns with the study by McKay et al,¹⁷ who reported a higher time-loss IID during the regular season compared to the playoffs. The heightened intensity and physicality of the playoffs in addition to the sheer number of games (16) required to win the Stanley Cup championship may translate to an increased risk of accidental collisions, intentional checks, and high-impact plays, and could predispose players to both acute traumatic injuries and fatigue-related overuse injuries. This increased risk during the playoffs is coupled with the cumulative physical toll of an 82-game NHL regular season. According to Pettigrew,²⁶ when comparing the total number of non-overtime hits during 8 NHL seasons, the hits per game during the playoffs increased 36.2% relative to regular season games. Not only does play during the playoffs become more intense and physical, which increases energy exertion, but also players may be relied on for extended minutes of play during both regulation time and overtime (which is longer in the playoffs). Higher workloads via increased minutes played per game have been associated with higher injury rates among professional athletes.^{5,9,13,14,18,30} However, this higher overall IID was not mirrored when examining time-loss injuries only. Our study found no significant association between injury mechanism and timing during season (ie, regular season vs playoffs) when examining time-loss injuries only. This finding along with the increased stakes of continuing to play during playoff games despite injury may contribute to the higher overall injury incidence not translating to time loss in the playoffs. This underscores the need for enhanced training strategies focused on injury reduction and fatigue management in professional hockey players, especially during the playoffs.

Injuries to the head (21.2%), hand (12.9%), hip/groin (9.6%), knee (8.7%), and shoulder (8.0%) were the most commonly injured body regions. Notably, knee injuries resulted in the highest mean games lost per injury (4.2), followed by shoulder (3.6), ankle (2.9), abdominal (2.7), and hip/groin (2.5) injuries. McKay et al¹⁷ similarly identified head injuries (16.8%) to be the most commonly injured body region, followed by the thigh (14.0%) and knee (13.0%). Unfortunately, these authors did not assess games lost per injured body region.¹⁷ In our investigation, concussions accounted for 3.9% of total injuries (18.1% of head injuries), resulting in a mean of 7.7 games lost. Although not officially affiliated with the NHL or NHL curated data, a cohort study analyzing the incidence, return to play, and performance of NHL players experiencing concussions from the 2009-2010 through 2015-2016 seasons similarly reported that players missed a mean of 7.5 games postconcussion.³ An epidemiological study (not officially affiliated with the NHL or NHL curated data) of NHL and Ontario Hockey League players from the 2009-2010 through 2011-2012 seasons comparing concussion incidence before and after a rule change making targeting an opponent's head from the "blind side" illegal reported no reduction in the number of players experiencing concussions despite this rule change.⁶ During the 2019-2020 NHL season, a new rule was established that assessed a minor penalty to a player whose helmet came off during play if the player did not exit the ice or retrieve and replace the helmet within a "reasonable period of time."¹⁰ These examples illustrate how the NHL evaluates and implements rule changes for competition based on epidemiological injury data. These changes can influence injury rates and the types of injuries in the athletes, with the goal of improving player safety.

TABLE 6
Predictors of Sustaining Injury Resulting in ≥ 5 Games Missed^a

	No. of MGL	No. of Injuries	No. of Injuries Resulting in ≥ 5 MGL	% Injuries Resulting in ≥ 5 MGL	OR	Lower 95% CI	Upper 95% CI	P Value	Significant
Time of season									
Preseason	4102	1649	189	11.5%	1.35	1.19	1.53	<.001	Yes
Regular season	46,580	21,077	2667	12.7%	1.57	1.43	1.71	<.001	Yes
Playoffs	1695	2050	95	4.6%	0.47	0.41	0.55	<.001	Yes
Position									
Forward	30,449	15,162	1727	11.4%	0.75	0.70	0.80	<.001	Yes
Defense	17,878	8642	990	11.5%	0.75	0.70	0.81	<.001	Yes
Goaltender	4050	972	234	24.1%	1.78	1.59	1.99	<.001	Yes
Body region									
Abdomen	1028	374	82	21.9%	2.08	1.63	2.65	<.001	Yes
Ankle	4309	1510	246	16.3%	2.00	1.73	2.32	<.001	Yes
Back	2040	918	92	10.0%	0.81	0.65	1.00	.05	No
Chest	1684	950	121	12.7%	1.02	0.84	1.24	.83	No
Foot	1646	1593	114	7.2%	1.19	0.97	1.47	.10	No
Head	8918	5259	490	9.3%	0.87	0.77	0.97	.01	Yes
Hand	4952	3189	298	9.3%	1.02	0.89	1.16	.79	No
Hip/groin	5986	2384	375	15.7%	1.24	1.09	1.42	<.001	Yes
Lower leg	1718	846	80	9.5%	1.27	1.01	1.60	.04	Yes
Knee	8972	2155	475	22.0%	2.13	1.90	2.39	<.001	Yes
Neck	1210	974	48	4.9%	0.32	0.24	0.42	<.001	Yes
Shoulder	7093	1994	362	18.2%	1.39	1.22	1.58	<.001	Yes
Thigh	1135	1100	73	6.6%	0.55	0.43	0.69	<.001	Yes
Upper arm	1368	1316	80	6.1%	0.58	0.47	0.73	<.001	Yes
Other	318	214	15	7.0%	0.58	0.35	0.96	.03	Yes
Game period									
Pregame warm-up	774	270	28	10.4%	0.82	0.58	1.16	.26	No
1st period	20,638	9470	1153	12.2%	1.12	0.99	1.26	.07	No
2nd period	16,155	7185	910	12.7%	1.18	1.04	1.33	.01	Yes
3rd period	12,415	6483	754	11.6%	1.07	0.94	1.21	.33	No
Overtime	473	191	26	13.6%	1.47	1.01	2.13	.04	Yes
Unknown	1922	1177	80	6.8%	0.71	0.61	0.83	<.001	Yes
Injury category									
Hit by puck	6977	6413	443	6.9%	0.52	0.46	0.58	<.001	Yes
Body check	13,738	4364	816	18.7%	1.81	1.66	1.99	<.001	Yes
Other intentional player contact	3593	2240	199	8.9%	0.87	0.76	1.00	.05	No
Incidental contact with another player or their equipment	7749	3858	429	11.1%	0.93	0.84	1.03	.16	No
Contact with environment	7292	2510	355	14.1%	1.24	1.11	1.38	<.001	Yes
Fighting	1453	785	87	11.1%	1.07	0.87	1.32	.51	No
Noncontact	7900	2392	443	18.5%	1.40	1.23	1.58	<.001	Yes
Unknown	3675	2214	179	8.1%	0.71	0.61	0.83	<.001	Yes

^aMGL, man-games lost.

Fighting-related injuries accounted for only 3.2% of all injuries in this analysis. Unfortunately, there is only anecdotal data regarding the actual risk for serious injuries resulting from fighting.²⁷ There has been a call for complete elimination of fighting in all amateur and professional hockey to reduce the risk of concussion and other potential musculoskeletal injuries.^{27,29} Fights per game in the NHL have diminished during the past 20 years. A study by Fortney et al⁸ found that fights per game decreased from a peak of 0.64 in 2002 to a low of 0.18 in 2020. A significant negative correlation was found between fights per game and fan attendance as well as between fights per game and goals per game.⁸ Therefore,

prohibiting fighting in the NHL could improve player safety without negatively impacting fan attendance. However, at present, the relatively declining injury risk resulting from fighting during this period of surveillance should provide some degree of optimism regarding the injury burden resulting from this aspect of the game.

This investigation is not without limitations. The retrospective analysis of prospectively collected data in this study precludes the establishment of causal relationships between variables. Additionally, we were limited to data reported in the AHMS, which is completed by team medical personnel based on their assessment of the injury mechanism and type. Particularly, injury types were categorized

by body region and predefined subcategories, limiting analysis for specific injuries (eg, shoulder labral tears, knee anterior cruciate ligament tears, etc) to the variables evaluated in this investigation. Clinician reporting practices within the constraints of the AHMS may vary, and return-to-play decisions based on medical criteria may differ based on the nature of the injury, experience/expertise of the clinicians, time of season, and other unknown factors that could not be quantified. Nevertheless, utilization of the AHMS is mandatory within the NHL, including its embedded injury surveillance, minimizing any potential bias from underreporting. Injuries sustained during practice were reported within the AHMS database; however, despite differences in practice frequency, duration, and intensity between teams, further investigations evaluating the incidence, types, and impact of practice-related injuries on time lost are warranted. Lastly, the COVID-19 pandemic occurred within the data collection period, potentially introducing bias into observed injury patterns, reporting practices, player susceptibility, player availability, and other unknown factors. This is especially possible in the seasons after the abbreviated 2019-2020 and 2020-2021 seasons. To mitigate this potential source of bias, we excluded the 2 seasons directly affected by the COVID-19 pandemic from our analysis. Unfortunately, the pandemic may have had indirect effects on injury rates and return to play after the excluded seasons that cannot be quantified.

Despite these limitations, this study is the largest comprehensive analysis of injuries sustained by professional hockey players in the NHL using a league-mandated injury database. Mandated use of a league-wide EMR has enabled reliable and consistent data collection and analysis for all NHL teams over the study period without the use of publicly available injury data retrieved through popular media sources. Accurate injury data are necessary to understand the burden, cause, and risk factors of injuries in the NHL and can help guide efforts to further improve the safety and health of these athletes.

CONCLUSION

From the 2012-2013 through 2022-2023 NHL seasons (excluding the 2019-2020 and 2020-2021 seasons shortened by the COVID-19 pandemic), the highest number of injuries occurred during the regular season, while the highest rate of injury occurred during the playoffs. The greatest incidence of injuries occurred in forwards, during the first period, due to puck-related causes. Injuries occurred most commonly to the head and hand, while injuries to the knee and shoulder accounted for the greatest mean number of man-games and days lost.

ACKNOWLEDGMENT

The authors thank the athletic trainers of the National Hockey League who were instrumental in entering all of

the injury data used for this study into the league's Athlete Health Management System.

ORCID iDs

Derrick M. Knapik  <https://orcid.org/0000-0001-8692-4746>

Darius Sina Tartibi  <https://orcid.org/0009-0005-6092-3776>

Robert H. Brophy  <https://orcid.org/0000-0002-2912-8265>

REFERENCES

1. Ajami S, Arab-Chadegani R. Barriers to implement electronic health records (EHRs). *Mater Sociomed*. 2013;25(3):213-215.
2. Anderson GR, Melugin HP, Stuart MJ. Epidemiology of injuries in ice hockey. *Sports Health*. 2019;11(6):514-519.
3. Andrews E, Jildeh TR, Abbas MJ, et al. Concussions in the National Hockey League: analysis of incidence, return to play, and performance. *Orthop J Sports Med*. 2022;10(1):23259671211052069.
4. Benson BW, Meeuwisse WH, Rizos J, Kang J, Burke CJ. A prospective study of concussions among National Hockey League players during regular season games: the NHL-NHLPA Concussion Program. *CMAJ*. 2011;183(8):905-911.
5. Blanch P, Gabbett TJ. Has the athlete trained enough to return to play safely? The acute:chronic workload ratio permits clinicians to quantify a player's risk of subsequent injury. *Br J Sports Med*. 2016;50(8):471-475.
6. Donaldson L, Asbridge M, Cusimano MD. Bodychecking rules and concussion in elite hockey. *PLoS One*. 2013;8(7):e69122.
7. Flik K, Lyman S, Marx RG. American collegiate men's ice hockey: an analysis of injuries. *Am J Sports Med*. 2005;33(2):183-187.
8. Fortney TA, Tedesco LJ, Kopydlowski NJ, et al. National Hockey League fights per game and viewership trends: 2000-2020. *Front Sports Act Living*. 2022;4:890429.
9. Freeston J, Soloff L, Schickendantz M, et al. In-game workload demands of position players in Major League Baseball. *Sports Health*. 2024;16(4):637-643.
10. Historical Rule Changes. NHL Records. Accessed April 13, 2024. <https://records.nhl.com/history/historical-rule-changes>
11. Huang C, Koppel R, McGreevey JD III, Craven CK, Schreiber R. Transitions from one electronic health record to another: challenges, pitfalls, and recommendations. *Appl Clin Inform*. 2020;11(5):742-754.
12. Hutchison MG, Di Battista AP, Meeuwisse W, et al. Concussion characteristics in the National Hockey League before and after the introduction of Rule 48. *JAMA Netw Open*. 2023;6(11):e2344399.
13. Lewis M. It's a hard-knock life: game load, fatigue, and injury risk in the National Basketball Association. *J Athl Train*. 2018;53(5):503-509.
14. Li RT, Salata MJ, Rambhia S, Sheehan J, Voos JE. Does overexertion correlate with increased injury? The relationship between player workload and soft tissue injury in professional American football players using wearable technology. *Sports Health*. 2020;12(1):66-73.
15. Lorentzon R, Wedren H, Pietila T. Incidence, nature, and causes of ice hockey injuries. A three-year prospective study of a Swedish elite ice hockey team. *Am J Sports Med*. 1988;16(4):392-396.
16. Martin C, Lieb A, Tokish J, et al. Initial versus subsequent injury and illness and temporal trends among professional hockey players. *Int J Sports Phys Ther*. 2024;19(2):215-226.
17. McKay CD, Tufts RJ, Shaffer B, Meeuwisse WH. The epidemiology of professional ice hockey injuries: a prospective report of six NHL seasons. *Br J Sports Med*. 2014;48(1):57-62.
18. Menon S, Morikawa L, Tummala SV, Buckner-Petty S, Chhabra A. The primary risk factors for season-ending injuries in professional basketball are minutes played per game and later season games. *Arthroscopy*. 2024;40(9):2468-2473.

19. Molsa J, Airaksinen O, Nasman O, Torstila I. Ice hockey injuries in Finland. A prospective epidemiologic study. *Am J Sports Med.* 1997;25(4):495-499.
20. Molsa J, Kujala U, Nasman O, Lehtipuu TP, Airaksinen O. Injury profile in ice hockey from the 1970s through the 1990s in Finland. *Am J Sports Med.* 2000;28(3):322-327.
21. *National Hockey League Official Rules 2013-2014.* National Hockey League; 2013.
22. *National Hockey League Official Rules 2023-2024.* National Hockey League; 2023.
23. NHL concussion evaluation and management protocol for 2022-23 season. NHL Public Relations. Accessed July 15, 2022. <https://www.nhl.com/news/nhl-concussion-evaluation-and-management-protocol-for-2022-23-season-335002568>
24. NHL Historical Rule Changes, 1910-2024. Scouting the Refs. Updated September 19, 2024. <https://scoutingtherefs.com/2023/04/38006/nhl-historical-rule-changes-1910-2023/>
25. Pettersson M, Lorentzon R. Ice hockey injuries: a 4-year prospective study of a Swedish elite ice hockey team. *Br J Sports Med.* 1993;27(4):251-254.
26. Pettigrew S. Playoff hockey is 36 percent more bone-crunching. Accessed April 13, 2024. <https://fivethirtyeight.com/features/play-off-hockey-is-36-percent-more-bone-crunching/>
27. Plassche GC, Fortney TA, Morrisette C, Korzelius JF, Popkin CA. Fighting in professional ice hockey: it's time for a change. *Phys Sportsmed.* 2023;51(5):405-413.
28. Sim FH, Simonet WT, Melton LJ III, Lehn TA. Ice hockey injuries. *Am J Sports Med.* 1987;15(1):30-40.
29. Smith AM, Stuart MJ, Dodick DW, et al. Ice Hockey Summit II: zero tolerance for head hits and fighting. *Clin J Sport Med.* 2015;25(2):78-87.
30. Soligard T, Schwellnus M, Alonso JM, et al. How much is too much? (Part 1) International Olympic Committee consensus statement on load in sport and risk of injury. *Br J Sports Med.* 2016;50(17):1030-1041.
31. Tuominen M, Stuart MJ, Aubry M, Kannus P, Parkkari J. Injuries in men's international ice hockey: a 7-year study of the International Ice Hockey Federation Adult World Championship Tournaments and Olympic Winter Games. *Br J Sports Med.* 2015;49(1):30-36.

Copyright of American Journal of Sports Medicine is the property of Sage Publications Inc. and its content may not be copied or emailed to multiple sites without the copyright holder's express written permission. Additionally, content may not be used with any artificial intelligence tools or machine learning technologies. However, users may print, download, or email articles for individual use.